An Ablation Study of Auxiliary Inputs for Lightweight Neural Upscaling in Real-Time Rendering

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Abstract

This manuscript is in preparation. The abstract will be written after the research question, experimental protocol, and results are finalized.

1 Introduction

Neural upscaling reconstructs a higher-resolution image from lower-resolution rendering data. A real-time renderer can also provide auxiliary signals, such as depth and motion vectors, that may help reconstruction. Each additional input has a potential quality benefit and a cost in data preparation, memory, and runtime. This study is planned as a controlled ablation of selected auxiliary inputs for a lightweight upscaler. The exact model, inputs, hardware, and evaluation conditions will be fixed in the experiment protocol before results are collected.

2 Background and Related Work

This section will compare relevant upscaling methods, rendering buffers, and evaluation practices.

3 Method

3.1 Research Questions and Study Design

The primary planned comparison is between an RGB-only baseline and variants with selected auxiliary inputs under a common model and evaluation setup. The final ablation matrix and its rationale are maintained in the experiment plan until the protocol is fixed.

3.2 Model, Inputs, and Data

The architecture, rendering inputs, and train/test scenes are to be selected.

3.3 Evaluation Protocol

The measurement protocol will be fixed before final experiments.

4 Results

Results have not yet been collected.

5 Discussion

This section will interpret quality and runtime tradeoffs after results are available.

6 Conclusion

Conclusions will be written after the analysis.

References